

Analysis and Modeling of Stress Using Wearable Sensor Data

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Course: STAT 355

Date: May 5, 2026

Introduction

For this study, I decided to use the “Nurse Stress Prediction Wearable Sensors” dataset from Kaggle. The dataset is composed of biometric data from nurses working in hospitals during the peak of the COVID-19 outbreak in the United States, which is approximately April 2020 to December 2020. The time frame is a period associated with increased workload and psychological stress. This context is important to consider when interpreting stress levels observed in the data.

Specifically, a few of the physiological signals tracked are electrodermal activity (EDA), heart rate (HR), and temperature (TEMP), along with labeled stress levels classified by nurses. My intention of this study is to investigate how physiological signals relate to stress levels. Understanding how these signals relate to stress, individually and in relation to one another, is important for contextualizing monitoring systems in healthcare settings.

Data Description

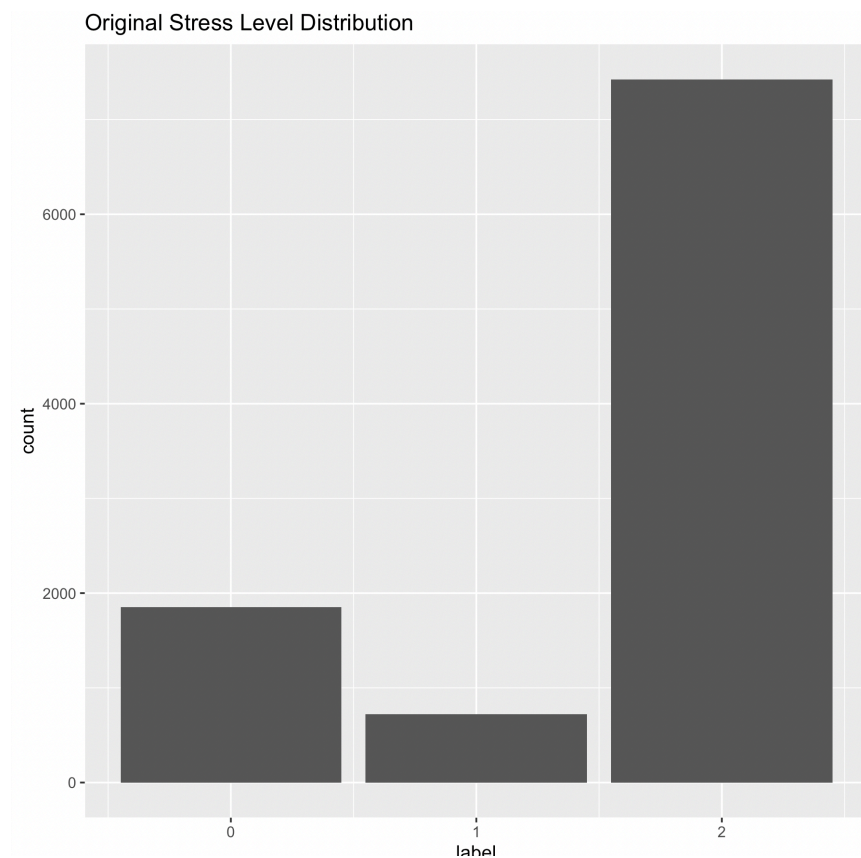
*Key Variables: *table on the next page**

Variable	Type
EDA: Electrodermal Activity –Measures skin conductance, which increases with activation of the sympathetic nervous	Continuous numerical
HR: Heart Rate –Represents the number of heartbeats per minute, which can increase with physical activity or stress.	Continuous numerical
TEMP: Body Temperature –Reflects skin or body temperature, which can vary due to environmental conditions and physiological regulation.	Continuous numerical
label: Stress Level -Categorical variable indicating stress level (0 = low, 1 = moderate, 2 = high), representing the observed or assigned stress state for each observation.	Categorical (discrete levels)

The dataset consists of approximately 11.5 million entries. I divided the data into 2 dataframes for efficiency and clear interpretation. I created **data_sample** to represent a random sample of 20,000 observations. I created **data_balanced** accounting for 2,000 observations per stress level (0,1,2) to address class imbalance and allow for fair comparisons.

Exploratory Data Analysis

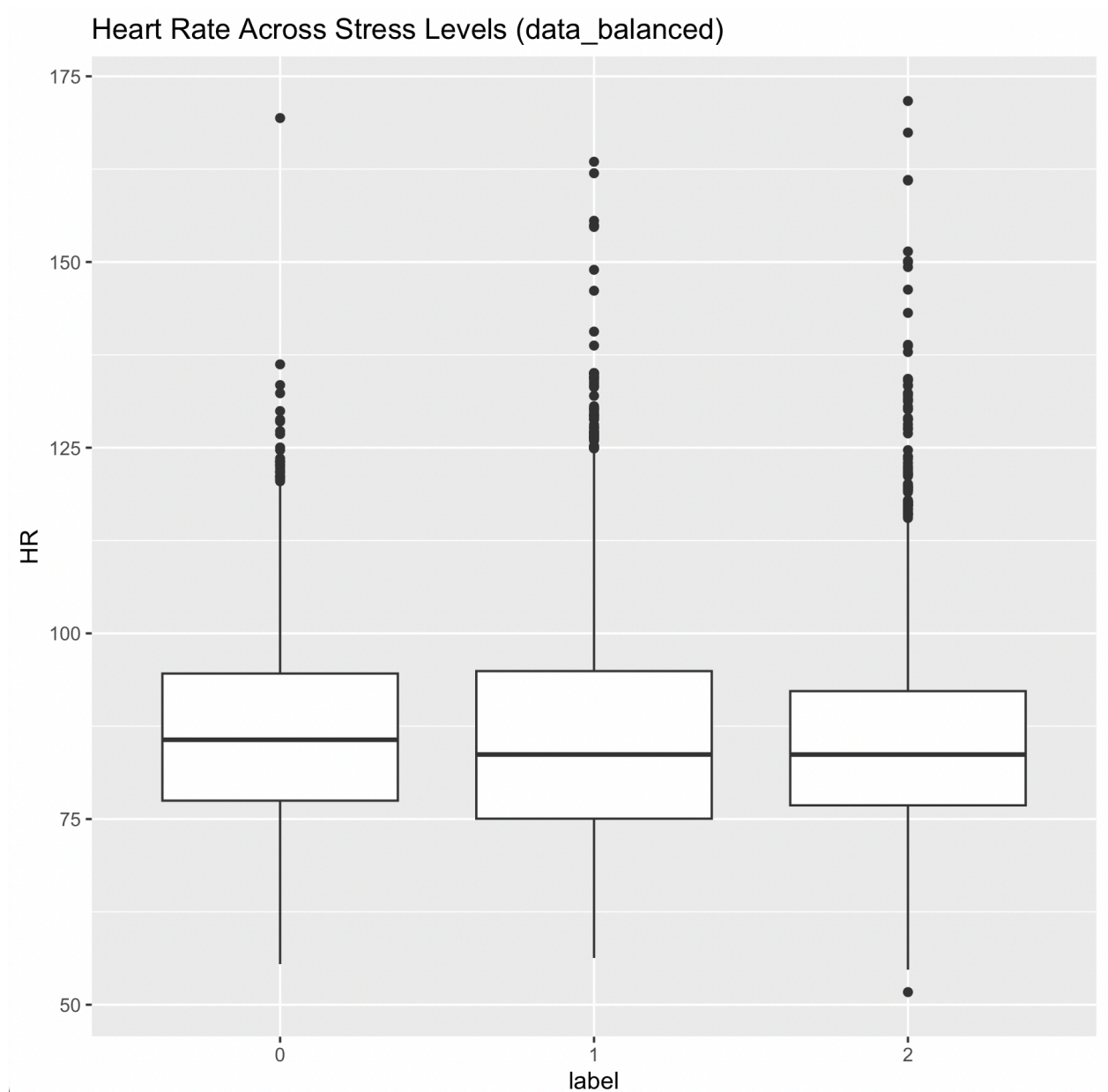
Stress Level Distribution: x-axis- 'label' & y-axis- 'count'



The distribution of stress levels for **data_sample** is highly imbalanced, with the majority of observations belonging to the highest stress category (label = 2). Moderate stress (label = 1) is significantly underrepresented, while low stress (label = 0) appears more frequently but still much less than high stress. This imbalance suggests that the dataset reflects a population with predominantly elevated stress levels, which may influence subsequent analyses.

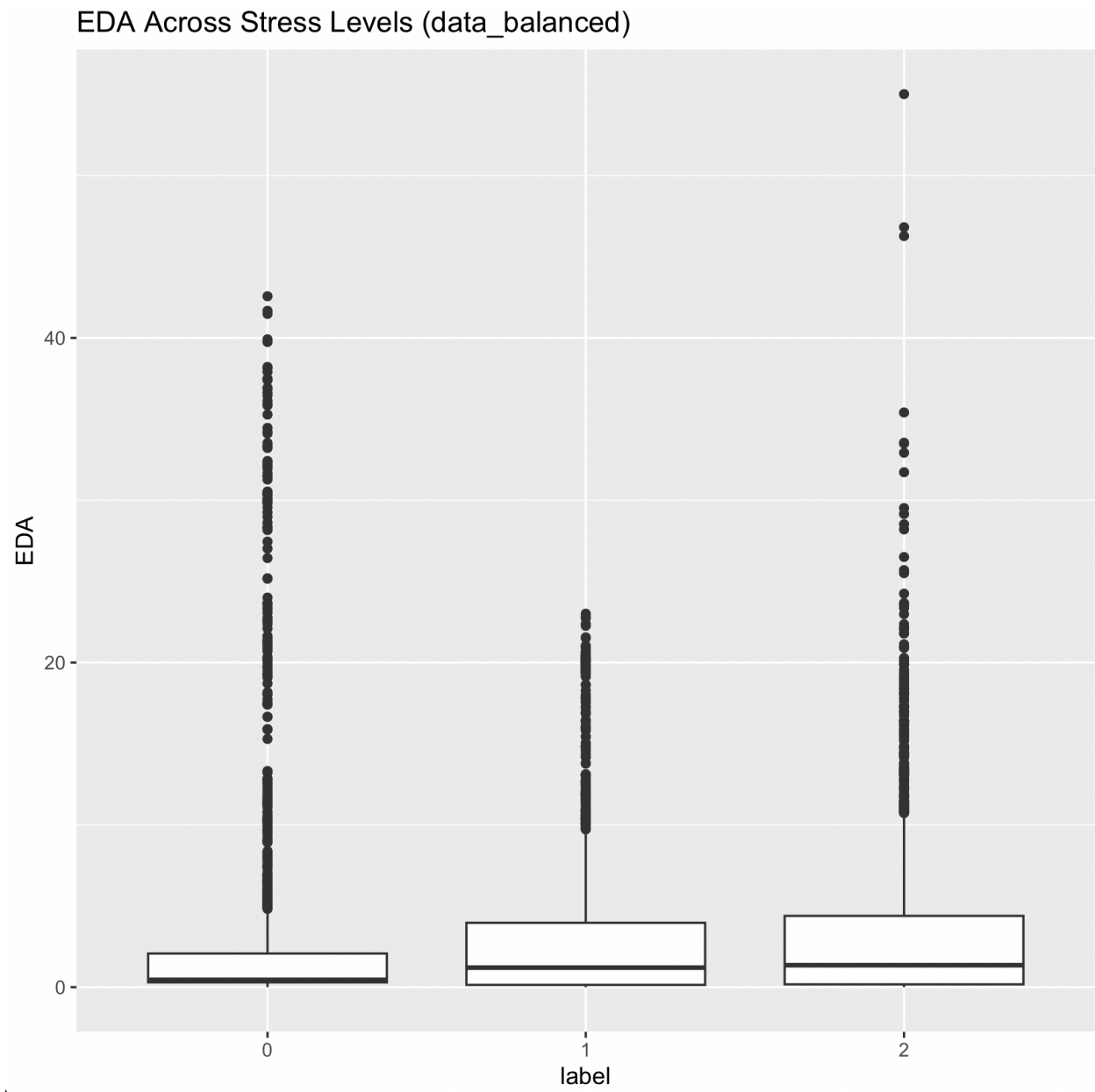
For modeling purposes, this imbalance leads to bias toward the majority class because the model learns to predict high stress in every case, which is not representative of the entire nurse population. To make sure each label is equally represented I implemented **data_balanced**, outlined above. Equal representation is important to classify data and stabilize hypothesis testing for interpretation.

Heart Rate vs Stress: x-axis- 'label' & y-axis- 'HR'



The boxplot of heart rate across stress levels show substantial overlap between all three groups. While there are minor differences in median heart rate, the overall distributions are quite similar, suggesting that heart rate alone may not strongly distinguish between stress levels in this dataset. The variability within each group is also relatively high, further indicating that heart rate may not be a reliable standalone predictor of stress.

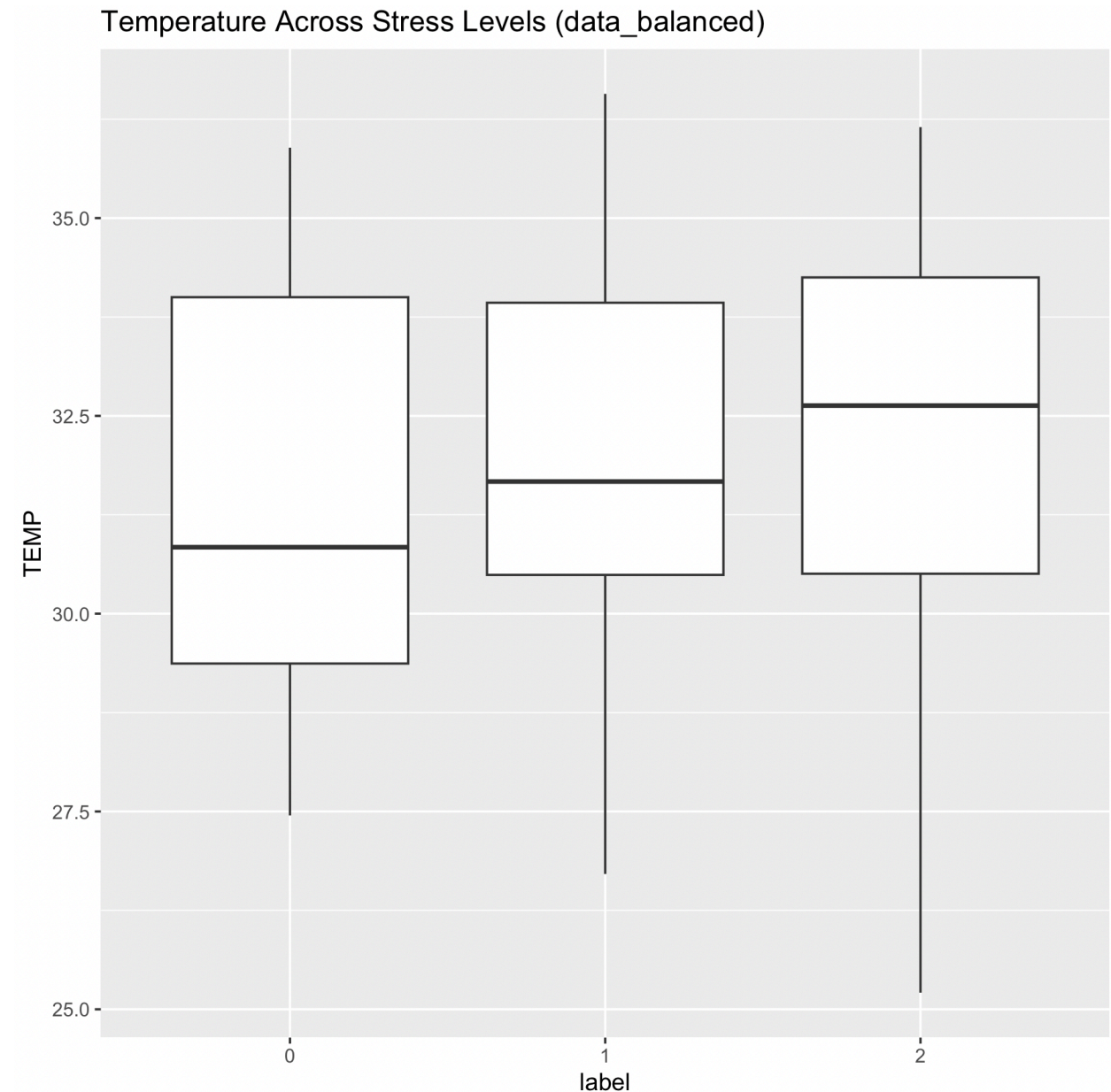
EDA Across Stress Levels: x-axis- 'label' & y-axis- 'EDA'



The boxplot of EDA across stress levels shows a clear upward trend, with median values increasing from low to high stress categories. Compared to heart rate, EDA exhibits more noticeable separations between groups, suggesting that it may be a more reliable physiological indicator of stress. Although there is still some overlap, the overall pattern indicates a positive relationship between EDA and stress level.

This aligns with physiological expectations, as electrodermal activity is directly linked to sympathetic nervous system activation, which increases during stress. Despite the overall trend, the presence of overlap and outliers suggests that EDA alone may not perfectly classify stress levels, indicating the potential benefit of combining multiple physiological signals.

Temperature Across Stress Levels: x-axis- 'label' & y-axis- 'TEMP'



The boxplot of temperature across stress levels shows a slight upward trend, with higher stress categories associated with marginally higher median temperatures. However, the substantial overlap between groups suggests that temperature is not a strong standalone indicator of stress. Compared to EDA, the relationship between temperature and stress appears weaker, though still present to some extent.

This may reflect the indirect relationship between stress and body temperature, which can be influenced by multiple physiological and environmental factors.

Correlation Matrix:

```
> cor(data_sample[, c("HR", "EDA", "TEMP")])
```

	HR	EDA	TEMP
HR	1.0000000	0.1419083	0.1517971
EDA	0.1419083	1.0000000	0.3500164
TEMP	0.1517971	0.3500164	1.0000000

To examine relationships between physiological variables, a correlation matrix was computed using heart rate (HR), electrodermal activity (EDA), and temperature (TEMP). The results indicate that HR has very weak correlations with both EDA ($r \approx 0.14$) and TEMP ($r \approx 0.15$), suggesting that heart rate does not strongly move in conjunction with the other signals. The strongest relationship observed was between EDA and TEMP ($r \approx 0.35$), which represents a weak to moderate positive correlation. Overall, these relatively low correlation values suggest that each physiological signal captures different aspects of the body's response rather than reflecting the same underlying pattern. This implies that no single variable is sufficient to explain the others, supporting the use of multiple features in later modeling and analysis.

Key Findings from Exploratory Data Analysis:

The exploratory data analysis revealed several patterns in the dataset that are essential to note before proceeding in research. I interpreted that the distribution of stress levels were highly imbalanced, with the majority of observations lying in the highest stress category. The imbalance of stress level ratings suggests that the dataset reflects a population experiencing a general trend of elevated stress levels, which should be carefully handled in further modeling of the dataset.

In terms of physiological signals, heart rate (HR) showed significant overlap across stress levels, indicating that it is not a strong predictor, on its own, affecting stress. In relation, electrodermal activity (EDA) demonstrated a clearer increasing trend as the stress levels increased, suggesting that it is more closely associated with stress responses than heart rate. Temperature (TEMP) showed a slight upward trend across stress levels, but the relationship was notably weaker and less consistent compared to EDA.

The correlation analysis table further supported my observations stated above, showcasing generally weak relationships between variables. This suggests that each physiological signal captures different aspects of the body's response to stress rather than reflecting a single underlying pattern.

Overall, the results indicate that while individual variables, particularly EDA, provide some insight into stress levels, no single feature fully explains stress levels by itself. This scientific observation supports the use of multiple variables in subsequent statistical and classification models.

Research Questions

Q1: Do Electrodermal Activity (EDA) levels differ significantly across stress levels?

Q2: How well does Electrodermal Activity (EDA) predict stress level?

Q3: Can physiological signals (EDA, heart rate, and temperature) be used to accurately predict stress level?

Q1- ANOVA Analysis

Model Set-Up:

Let μ_0 , μ_1 , μ_2 be the mean EDA for stress levels 0, 1, 2 (sequentially)

- H_0 : null hypothesis- $\mu_0 = \mu_1 = \mu_2$
-> mean EDA is the same across stress levels
- H_1 : alternative hypothesis- at least one mean differs
EDA varies across stress levels ->

ANOVA analysis is used in this research question to compare the mean Electrodermal Activity (EDA) across multiple stress level groups (0,1, and 2) to determine whether there is a statistically significant difference between EDA by stress level.

ANOVA Output:

```
> anova_model <- aov(EDA ~ label, data = data_balanced)
> summary(anova_model)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
label	2	295	147.74	4.638	0.00971 **
Residuals	5997	191021	31.85		

```
---
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
> |
```

The ANOVA results show $F(2, 5997) = 4.638$ with $p = 0.00971$, so we reject the null hypothesis and conclude that there is a statistically significant difference in mean EDA across the three stress levels.

A Tukey post-hoc test was conducted to determine which stress levels differed in mean EDA.

Tukey Output:

```
> TukeyHSD(anova_model)
Tukey multiple comparisons of means
95% family-wise confidence level

Fit: aov(formula = EDA ~ label, data = data_balanced)

$label
      diff      lwr      upr    p adj
1-0 0.3012140 -0.1171785 0.7196064 0.2099481
2-0 0.5424813  0.1240889 0.9608738 0.0067243
2-1 0.2412673 -0.1771251 0.6596598 0.3665698
```

The results indicate that stress level 2 has significantly higher EDA compared to stress level 0 ($p \approx 0.0067$). However, no statistically significant differences were found between stress levels 1 and 0 or between 2 and 1. This suggests that the primary distinction in EDA occurs between low and high stress levels. This finding aligns with the exploratory analysis, where EDA appears to increase with stress level, though the separation between adjacent levels was less pronounced.

Interpretation:

The output supports that electrodermal activity (EDA) levels are not the same across all stress groups, suggesting that EDA increases as stress levels increase. To reiterate, EDA is related to stress and can help distinguish between different levels of stress, although it may not perfectly separate them on its own.

Q2- Linear Regression

Model Set-Up:

I am modeling stress as a function of EDA. The dependent y-variable is *label* (stress level) and the independent x-variable is *EDA*. The *label* is being treated as numeric in this case because the quantities within the category (0,1,2) are ordered based on the amount of stress. The key idea is that we are approximating an ordinal variable as numeric, to capture the model trend and test if stress increases with EDA.

Regression is used to examine the relationship between the two variables. Specifically, we are focused on if EDA predicts stress level and if there is a linear relationship between the two variables. The model is a simple linear regression.

It is represented as: $\text{Stress Level} = \beta_0 + \beta_1(\text{EDA})$

- β_0 : the intercept; predicted stress level when EDA = 0
- β_1 : the slope; how much the stress level is expected to change for a 1-unit change in EDA

Linear Regression Output:

```
> lm_model <- lm(label ~ EDA, data = data_sample)
> summary(lm_model)

Call:
lm(formula = label ~ EDA, data = data_sample)

Residuals:
    Min       1Q   Median       3Q      Max
-1.9149 -0.5261  0.4450  0.4712  0.4748

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.5252206   0.0041314   369.2  <2e-16 ***
EDA           0.0089588   0.0006266    14.3  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.785 on 49998 degrees of freedom
Multiple R-squared:  0.004072, Adjusted R-squared:  0.004052
F-statistic: 204.4 on 1 and 49998 DF, p-value: < 2.2e-16
```

The linear regression results indicate that electrodermal activity (EDA) is a statistically significant predictor of stress level. The estimated coefficient for EDA is $\beta_1 = 0.00896$ with a p-value $< 2e - 16$, indicating strong statistical significance. This suggests that as EDA increases, the predicted stress level also increases.

The model has an R^2 value of approximately 0.004, indicating that EDA explains only a very small portion of the variability in stress level. Although the relationship is statistically significant, the low R^2 suggests that EDA alone has limited predictive power for stress.

Interpretation:

The results indicate that while there is a statistically significant relationship between EDA and stress level, the strength of this relationship is weak. The positive coefficient suggests that higher EDA values are associated with higher stress levels, which aligns with physiological expectations. However, the very low R^2 value indicates that EDA alone does not explain much of the variation in stress levels.

This suggests that although EDA is related to stress, it is not a strong predictor on its own. The overlap observed in the exploratory analysis is reflected in the regression results above, indicating that additional variables are likely needed to more accurately model stress levels.

Q3- Classification Model

Model Set-Up:

To address this research question, a multinomial logistic regression model was used to classify stress levels based on physiological signals. The dependent variable is the categorical stress label (0 = low, 1 = moderate, 2 = high), and the independent variables include electrodermal activity (EDA), heart rate (HR), and temperature (TEMP). Multinomial logistic regression is appropriate in this case because the outcome variable consists of more than two categories. The model estimates the probability of each stress level based on the input features and assigns the most likely class as the prediction. This allows for evaluating how well the combination of physiological signals can be used to classify stress levels

*Classification Output using **data_sample***

```
> data_sample$label <- as.factor(data_sample$label)
> set.seed(123)
>
> train_index <- sample(1:nrow(data_sample), 0.7 * nrow(data_sample))
>
> train <- data_sample[train_index, ]
> test <- data_sample[-train_index, ]
> library(nnet)
>
> model <- multinom(label ~ EDA + HR + TEMP, data = train)
# weights: 15 (8 variable)
initial value 38451.430103
iter 10 value 27037.400798
iter 20 value 24849.664889
final value 24849.595230
converged
> predictions <- predict(model, newdata = test)
> mean(predictions == test$label)
[1] 0.7410667
> table(Predicted = predictions, Actual = test$label)
      Actual
Predicted    0     1     2
      0      0      0      0
      1      0      0      0
      2 2802 1082 11116
> |
```

The multinomial logistic regression model was first trained using the original sampled dataset (**data_sample**), which retains the natural class imbalance present in the data. The model achieved an overall accuracy of approximately 74%. However, the confusion matrix shows that the model predicts nearly all observations as the highest stress level (label = 2). This indicates that the model is biased toward the majority class due to the imbalance in the dataset. As a result, the model performs poorly in identifying low and moderate stress levels.

This demonstrates that although the model appears to perform well based on accuracy, it is not effectively distinguishing between stress categories. The high accuracy is therefore misleading, as it reflects the underlying class imbalance rather than true predictive performance.

*Classification Output using **data_balanced***

```
> set.seed(123)
>
> train_index <- sample(1:nrow(data_balanced), 0.7 * nrow(data_balanced))
>
> train_b <- data_balanced[train_index, ]
> test_b <- data_balanced[-train_index, ]
>
> model_b <- multinom(label ~ EDA + HR + TEMP, data = train_b)
# weights: 15 (8 variable)
initial value 4614.171612
iter 10 value 4560.647681
final value 4560.639824
converged
>
> pred_b <- predict(model_b, newdata = test_b)
>
> mean(pred_b == test_b$label)
[1] 0.3933333
> table(Predicted = pred_b, Actual = test_b$label)
      Actual
Predicted  0    1    2
      0 322 263 216
      1  27  73  83
      2 250 253 313
> |
```

The multinomial logistic regression model was then trained using the balanced dataset (**data_balanced**), where each stress level was equally represented. The model achieved an overall accuracy of approximately 39%.

The confusion matrix shows that the model predicts all three stress levels, rather than defaulting to a single dominant class. While the overall accuracy is lower compared to the model trained to the original sampled dataset (**data_sample**), the predictions are more evenly distributed across stress categories.

This indicates that the balanced dataset allows the model to better distinguish between different stress levels. Although performance appears worse based on accuracy alone, the model provides a more meaningful evaluation of classification ability because it is not biased toward a majority class. This demonstrates that balancing the dataset improves the model’s ability to capture patterns across all stress levels.

Side-by-Side Comparison:

Model	Dataset	Accuracy	Behavior
Original	data_sample	~74%	Predicts majority class
Balanced	data_balanced	~39%	Predicts all classes

Interpretation:

The comparison between the models trained on the original and balanced datasets highlights the impact of class imbalance on classification performance. The model trained on the original dataset achieved a higher accuracy; however, this result was misleading because the model primarily predicted the majority class (high stress) As a result, it failed to effectively distinguish between low and moderate stress levels.

In contrast, the model trained on the balanced dataset achieved a lower overall accuracy but demonstrated improved performance in predicting all stress categories. This indicates that the model was better able to learn patterns associated with each stress level when the data was evenly distributed.

Overall, these results demonstrate that accuracy alone is not a sufficient metric for evaluating classification models, particularly in the presence of class imbalance. The balanced model provides a more meaningful assessment of predictive performance, as it reflects the model’s ability to classify all stress levels rather than defaulting to a single dominant category.

Integrated Discussion

The results across all three analyses provide a consistent picture of the relationship between physiological signals and stress levels. The ANOVA results showed that mean EDA differs significantly across stress groups, indicating that EDA is associated with stress at a group level. This was supported by the exploratory analysis, where EDA displayed a clear increasing trend across stress levels.

However, the regression analysis revealed that although EDA is statistically significant, it has very low explanatory power when predicting stress at the individual level. This suggests that while EDA differs across groups it does not strongly predict stress on its own. This distinction highlights that group-level differences do not necessarily translate into strong predictive performance.

The classification results further reinforce this conclusion. Even when combining multiple physiological signals (EDA, HR, and TEMP), the model struggled to accurately classify stress levels, particularly when accounting for class imbalance. While balancing the dataset improved the model's ability to distinguish between classes, overall predictive performance remained limited.

These findings suggest that stress is a complex physiological state that cannot be fully captured by a single signal or even a small set of variables. Instead, more advanced modeling approaches or additional features may be necessary to improve predictive accuracy.

Limitations

Several limitations should be considered when interpreting the results of this analysis. First, the dataset exhibits significant class imbalance, which affects model performance and evaluation, particularly in classification tasks. Although a balanced dataset was used to address this issue, it may not fully represent the original data distribution.

Second, the physiological signals used in this study are inherently noisy and influenced by multiple factors beyond stress, such as physical activity and environmental conditions. This contributes to the overlap observed across stress levels and limits predictive performance.

Additionally, the dataset likely contains repeated measurements from the same individuals over time, which may violate the assumption of independence in statistical tests. This could impact the validity of some results.

Conclusion

In this study, the relationship between physiological signals and stress levels was analyzed using exploratory data analysis, ANOVA, linear regression, and classification modeling. The results showed that electrodermal activity (EDA) is the most informative individual variable out of the other ones, with significant differences observed across stress levels. However, regression and classification analyses demonstrated that EDA alone is not a strong predictor of stress.

Overall, the findings indicate that while physiological signals provide some insight into stress, they are insufficient on their own to accurately predict stress levels. This highlights the complexity of stress as a physiological phenomenon and suggests that more comprehensive models or additional data may be required for improved prediction.

Future Directions

Future work could expand this analysis by incorporating additional physiological and contextual variables such as physical activity, sleep quality, or time-based behavioral patterns. Since stress is dynamic and changes over time, future studies may also benefit from time-series modeling approaches

rather than relying on individual observations. Additionally, more advanced machine learning models and feature engineering techniques could improve classification performance and better capture the complexity of physiological stress responses. Finally, future research could explore the development of real-time wearable monitoring systems for stress detection in healthcare environments.